ROOTFORCE

WAZUH & SURICATA INTEGRATION

WAZUH

SURICATA

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Suricata and Wazuh Integration Documentation

1. Introduction

This document provides a comprehensive guide for integrating Suricata, an open-source Network Intrusion Detection System (NIDS), with Wazuh, a free and open-source security monitoring platform. The integration focuses on deploying Suricata on a Windows Server 2022 endpoint and forwarding its generated security events to a Wazuh Manager deployed as an OVF on VMware. This integration enhances an organization's security posture by combining Suricata's deep packet inspection capabilities with Wazuh's extensive security analytics, log management, and alerting features.

Suricata functions as a powerful NIDS, IPS, and Network Security Monitoring (NSM) engine, capable of real-time threat detection through signature-based and anomaly-based analysis of network traffic . Wazuh, on the other hand, offers Security Information and Event Management (SIEM) capabilities, including log data analysis, file integrity monitoring, vulnerability detection, and incident response . By integrating these two systems, security teams can gain deeper visibility into network threats and centralize their security event management.

2. Architecture Overview

The integration architecture involves Suricata running on a Windows Server 2022 endpoint, capturing and analyzing network traffic. The Suricata-generated alerts and events, primarily in eve.json format, are then collected by the Wazuh agent installed on the same Windows server. The Wazuh agent securely forwards these logs to the Wazuh Manager, which is deployed as an OVF appliance on a VMware environment. The Wazuh Manager processes these logs, correlates them with other security data, and makes them available for analysis and visualization in the Wazuh Dashboard.

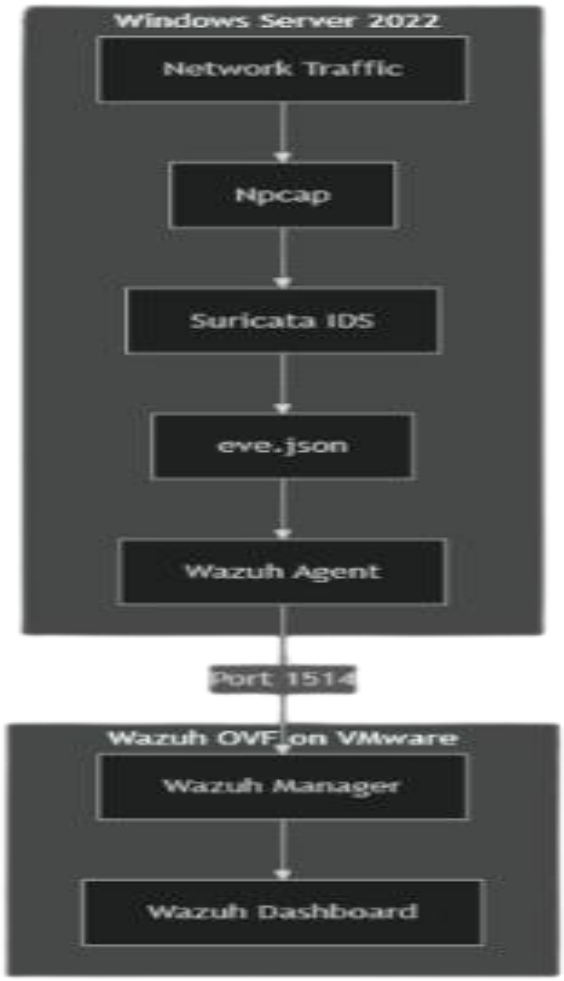

Figure 1: Suricata and Wazuh Integration Architecture

3. Prerequisites

Before proceeding with the integration, ensure the following prerequisites are met:

1•Windows Server 2022 or any windows OS: A running instance of Windows administrative access.

2. •Wazuh Manager: A Wazuh Manager deployed as an OVF appliance on VMware, fully operational and accessible from the Windows endpoint.
3. •Wazuh Agent: The Wazuh agent installed on the Windows endpoint and successfully reporting to the Wazuh Manager.
4. •Network Connectivity: Ensure proper network connectivity between the Windows and the Wazuh Manager
5. •Npcap/WinPcap: A packet capture library (Npcap recommended) for Suricata to capture network traffic.
6. Kali linux to do attack on the windows server to see how it look like in monitoring solutions

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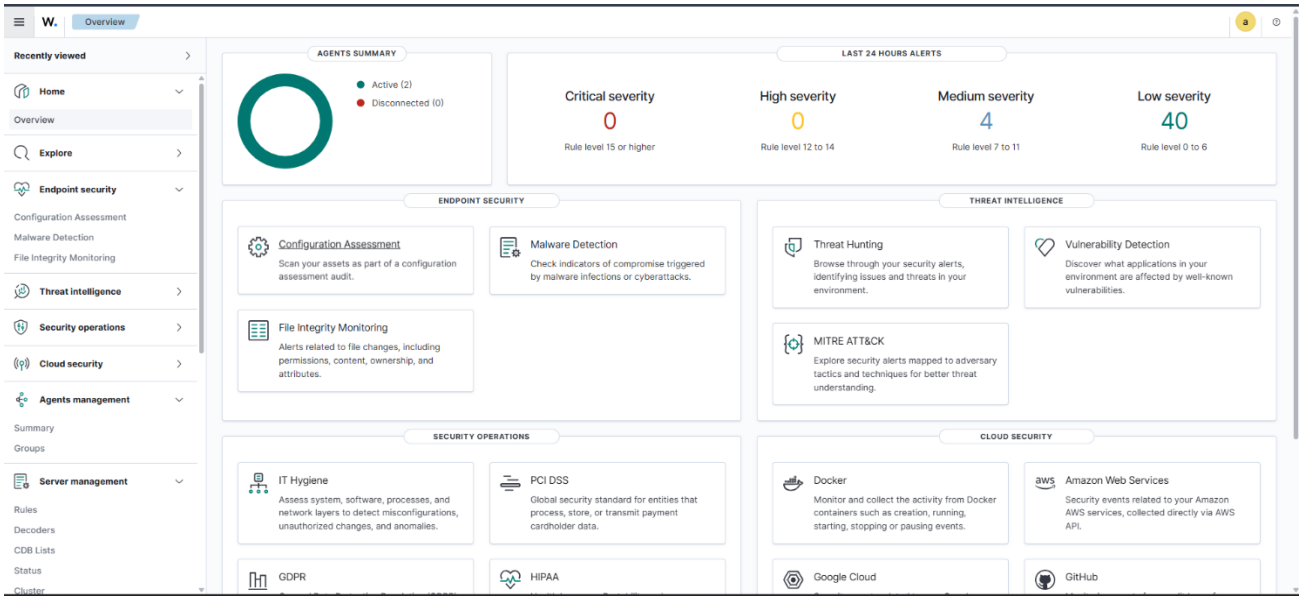
Setup Guide:

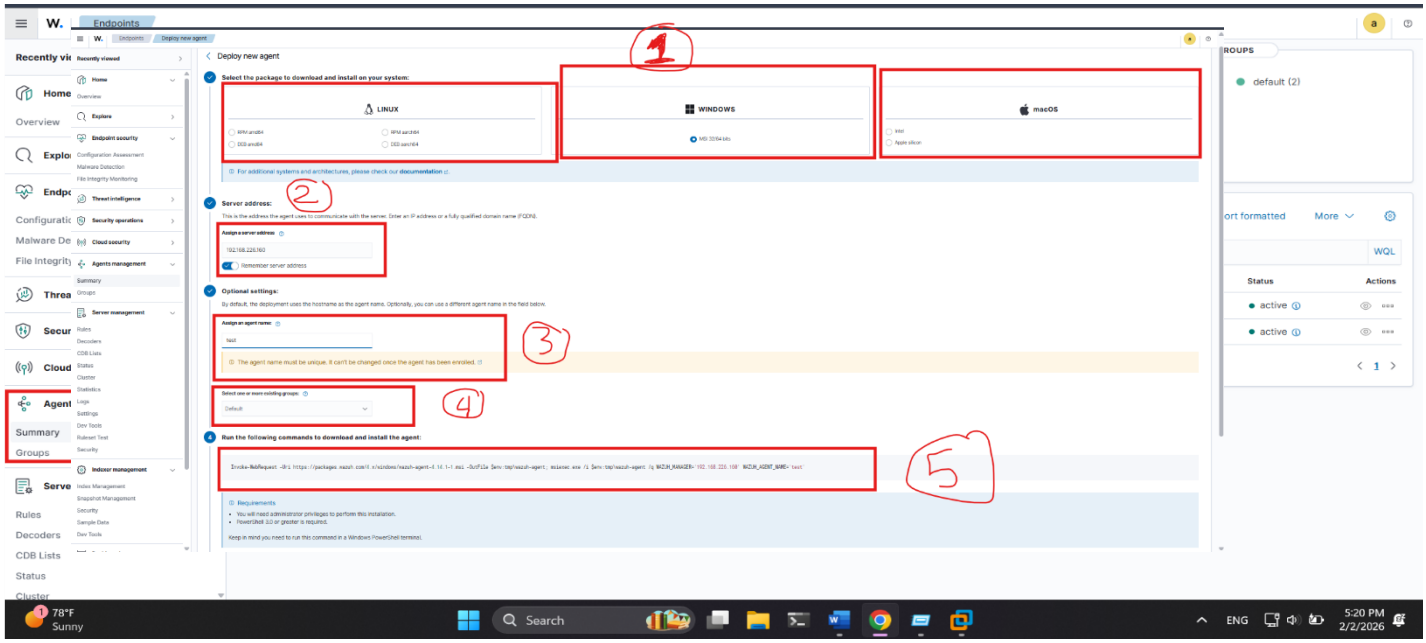
First lets have our Wazuh manager to be appear like this :



-note: that first time when you install the wazuh it will not show any active or disconnected devices.

After the Wazuh manager has been done and start working , lets install the agents on devices we want to monitor:

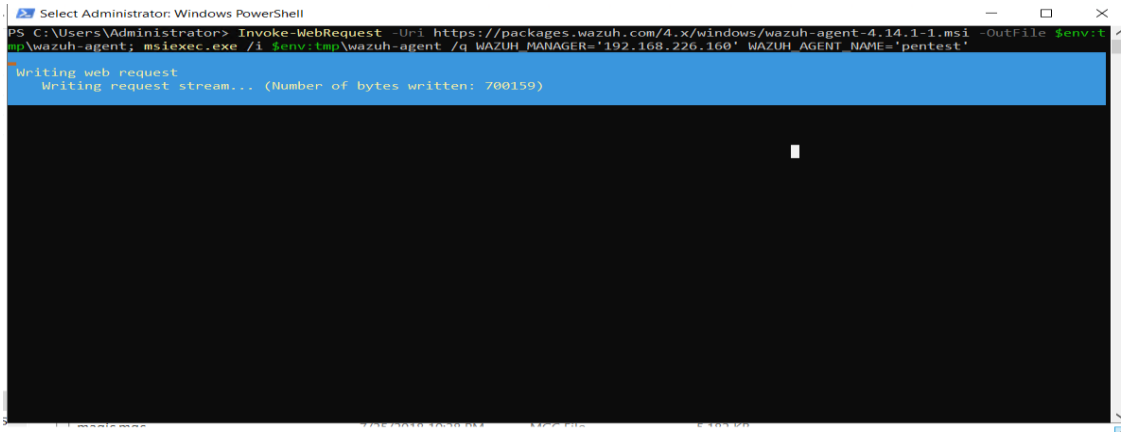
- ⇨ Dashboard of deploying new agents:



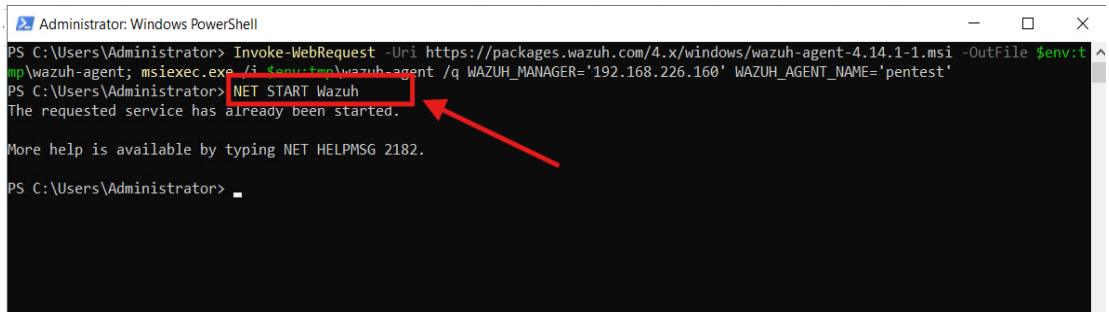
in the previous image there were(5) important sections we need to understand:

1. The type of OS you want to deploy your agent (windows, macOS, Linux)
2. Here you put the IP of Wazuh server (Wazuh manager)
3. Here you just put the name of agent that will appear in your dashboard lately and note you can not change any of these configurations after you finish deploying agent,
4. so focus
5. Here you choose the group of agents you want to put your agent in, means if you many department in your firm and you want to separate agents of every department
6. In this step is the command that you are going to run in OS you want to deploy the agent in, whether in power shell , command prompt , etc.

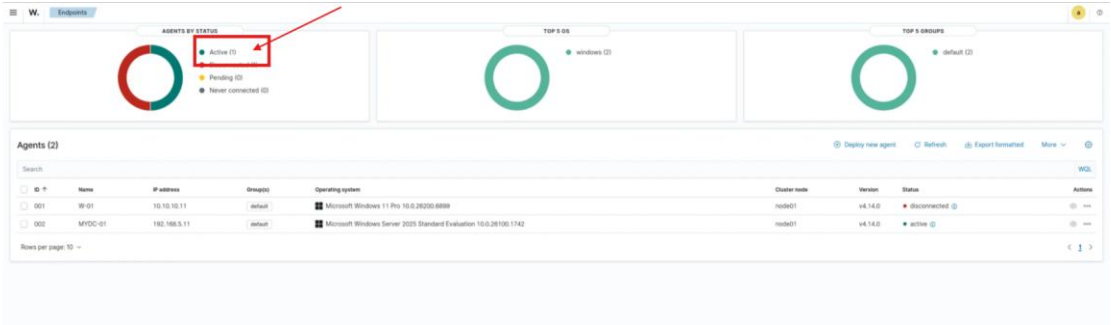
⇨ Installing Wazuh-Agent in windows:



Then start the Wazuh-Agent with following command :



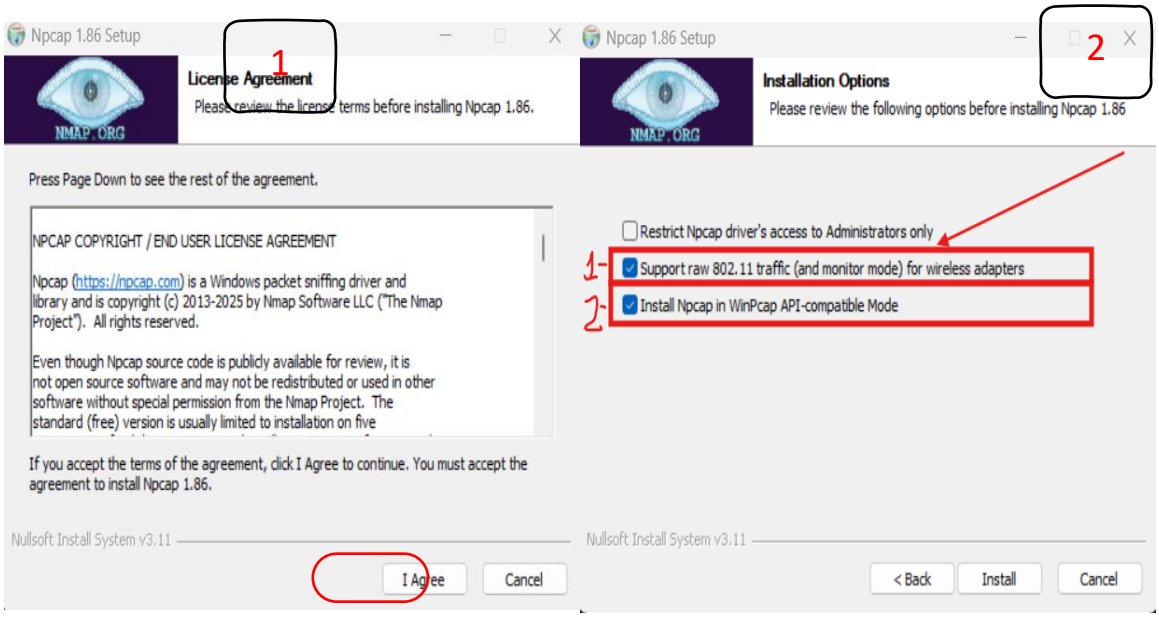
After wazuh-agent is installed it will appear in dashboard as active agent



Installing Suricata on windows:

Before installing Suricata we need to install Network traffic capturing software. I have installed npcap. Download Npcap from its official Website for windows: https://npcap.com/#download

Follow the instruction to install Npcap:



Ensure that option that are highlighted is:

Option -1- is necessary for allowing for capturing the wifi packet with npcap.

Option -2- is used for specifying and installing the npcan in the winpcap which is the packet capture.

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Overview of npcap and winpcap:

1. WinPcap (The Original)

WinPcap was the industry-standard "packet capture" library for Windows for nearly two decades.

What it does: Usually, Windows only shows an application the data specifically meant for it. WinPcap allows applications to bypass the standard protocol stack to "sniff" all data passing through the network card.

The Downside: It is now officially deprecated. It hasn't been updated since 2013, it doesn't support modern Windows 10/11 security features, and it can be unstable on newer systems.

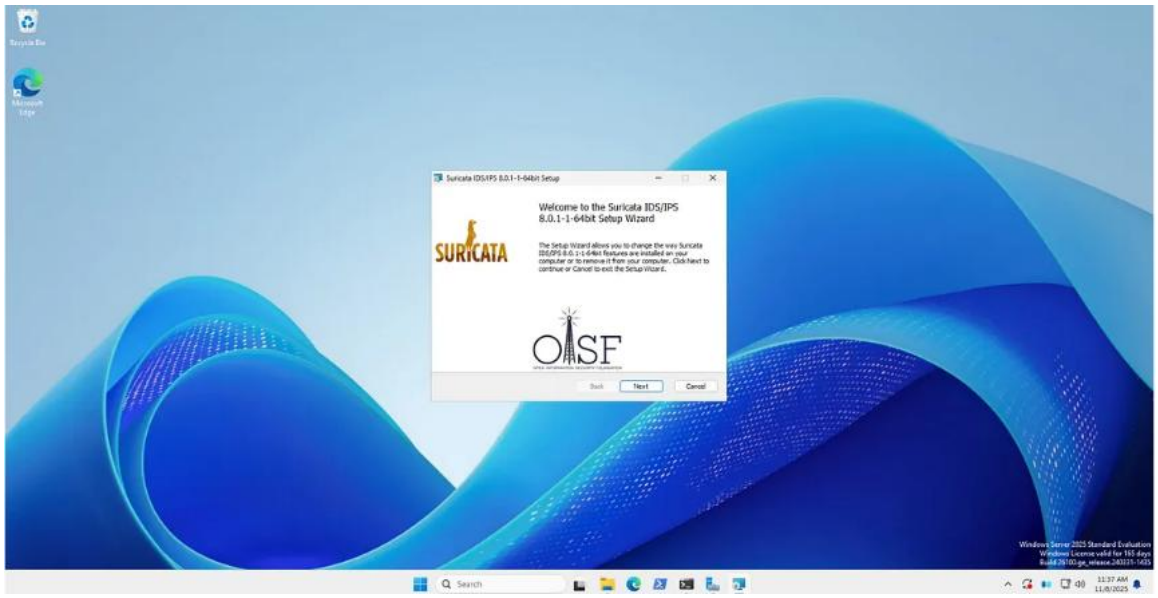
2. Npcap (The Modern Successor)

Npcap is the modern version of WinPcap, maintained by the Nmap project. It’s faster, more secure, and designed specifically for modern versions of Windows.

Installing Suricata in windows:

Download Suricata from its official website for Windows:

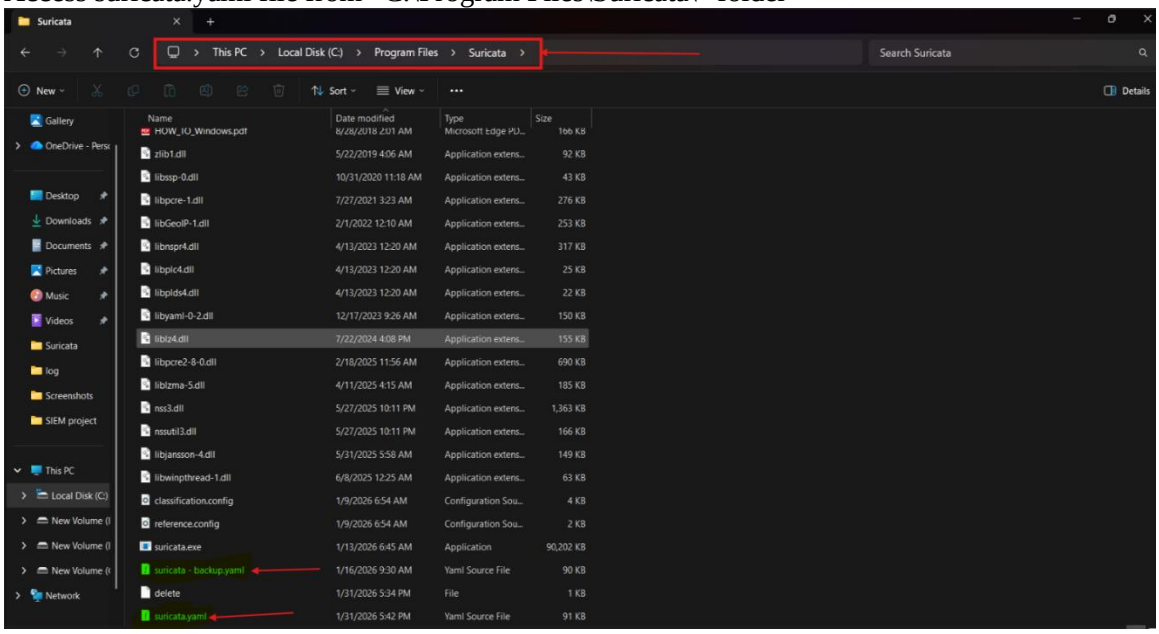
https://suricata.io/download/



Configuring Suricata:

suricata.yaml is the configuration file of suricata.

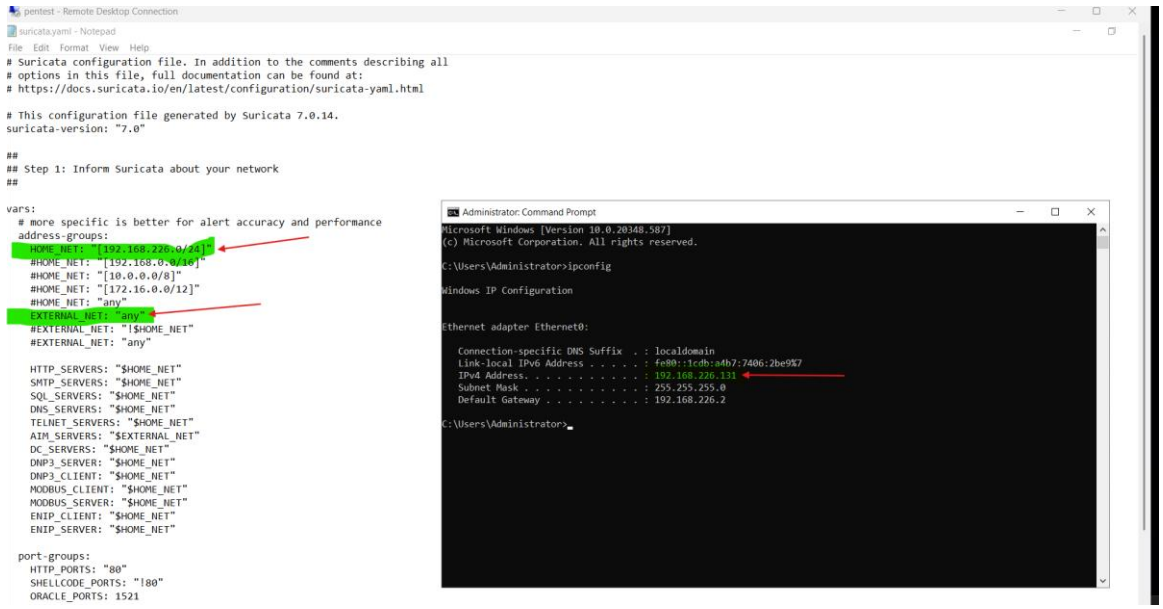
Access suricata.yaml file from “C:\Program Files\Suricata\” folder



Tip: And in here I want to mention something is very important , when editing any configuration file you should first have a backup of that configuration .Because this will help in fixing many problems if you miss a lot with configuration file.

-know let edit the suricata.yaml which is the configuration file of suricata:

Changing HOME_NET to the network id of my server Ip address and allow for EXTERNAL_NET for allowing the integration with Wazuh server:

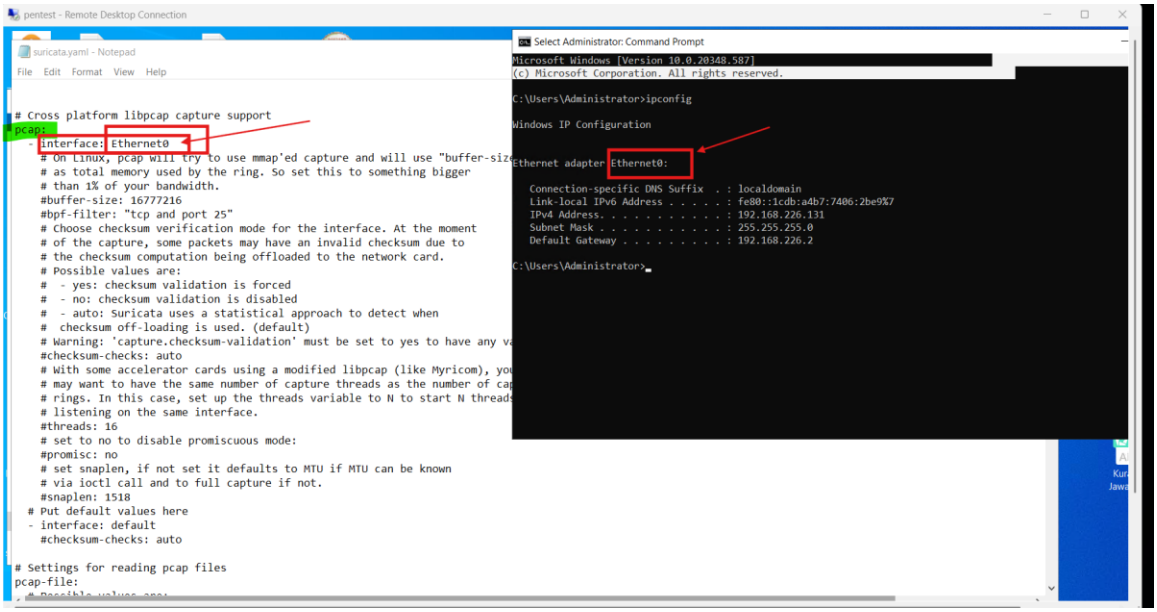


Network Interface Binding (Suricata + Windows)

The following image demonstrates the critical step of aligning Suricata's packet capture engine with the server's physical or virtual hardware.

- **Interface Identification:** The Command Prompt window shows the output of `ipconfig`, identifying the active network adapter as **"Ethernet0"**. This is the interface currently assigned the IP address `192.168.226.131`.
- **Suricata YAML Configuration:** The Notepad window shows the `suricata.yaml` configuration file. Specifically, under the `pcap` (Packet Capture) section, the `interface` parameter has been manually updated to **"Ethernet0"** to match the system's adapter name.
- **Operational Significance:** This binding ensures that Suricata is "listening" to the correct stream of network traffic. Without this precise match, the IDS/IPS engine will fail to initialize or will monitor an inactive interface, resulting in zero threat detection.

⇨ **Configuring the Listening Interface**



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⇨ Configuring the rules in Suricata:

Download the rules of emerging threats from Proofpoint Emerging Threats.

Link : https://rules.emergingthreats.net/

⇨ Note: the rules must match the version of IDS Suricata or Suricata will show error.

Proofpoint Emerging Threats Rules

Please review the instructions for Pro and Open rule downloads.

Name	Last Modified	Size
OPEN_download_instructions.html	2025-02-19T19:27:59Z	5.92 KB
PRO_download_instructions.html	2025-02-19T19:27:59Z	6.18 KB
blockrules/	2026-01-30T05:30:04Z	-
changelogs/	2026-01-30T21:17:54Z	-
fwrules/	2014-08-11T17:22:20Z	-
open/	2026-01-30T21:15:19Z	-
open-nogpl/	2026-01-30T21:15:19Z	-
version.txt	2026-01-30T21:15:11Z	6 B

Support: Discourse site, Twitter, Email

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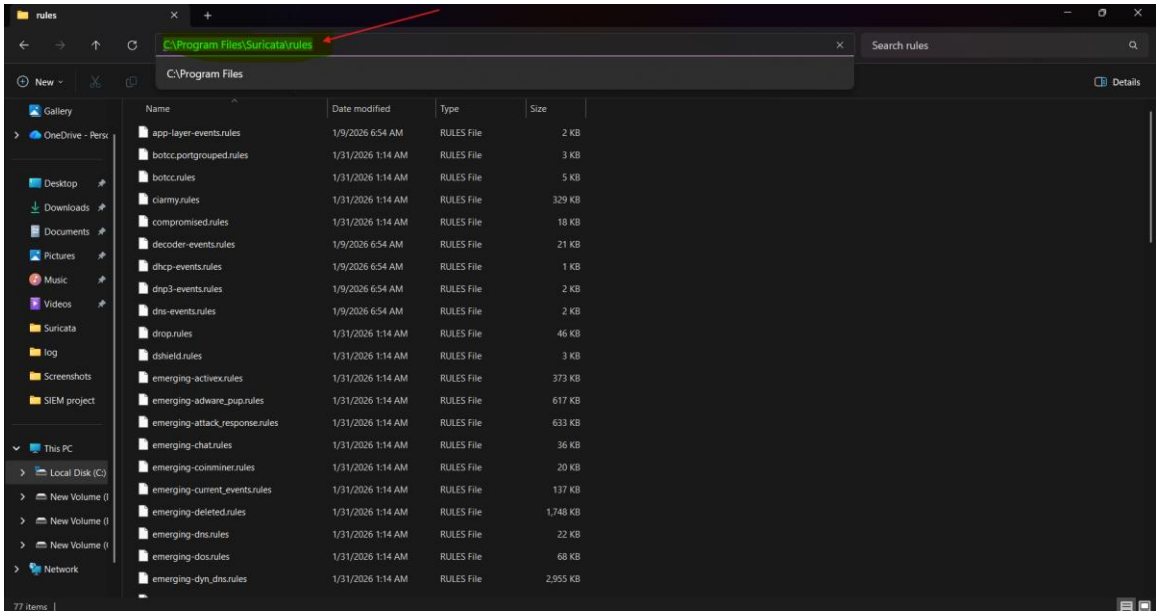
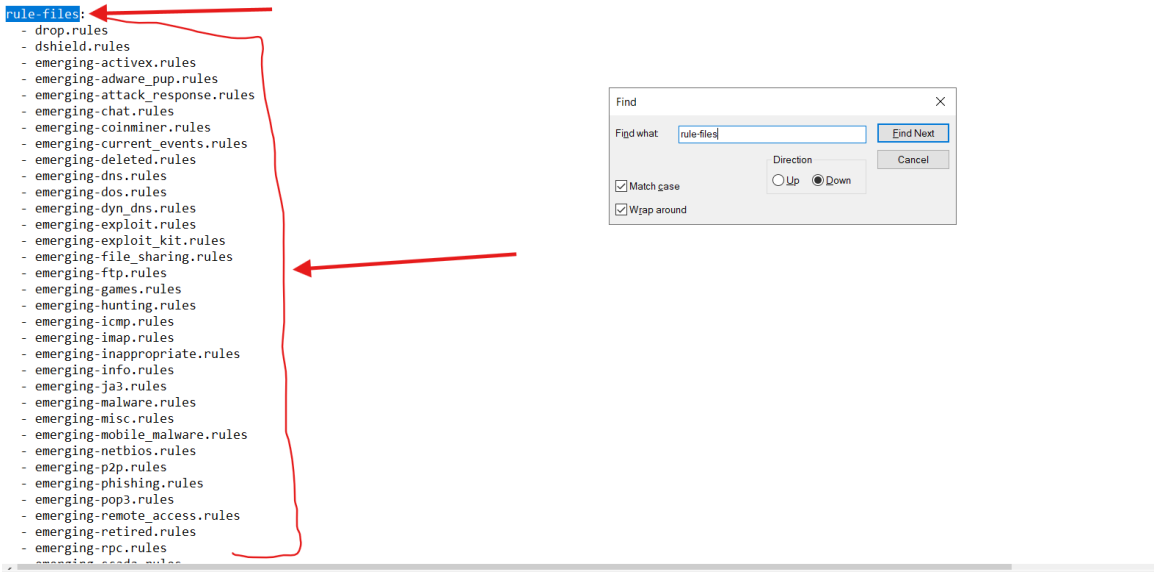
Extract the rules and copy it to “C:\Program Files\Suricata\rules“ folder like following:

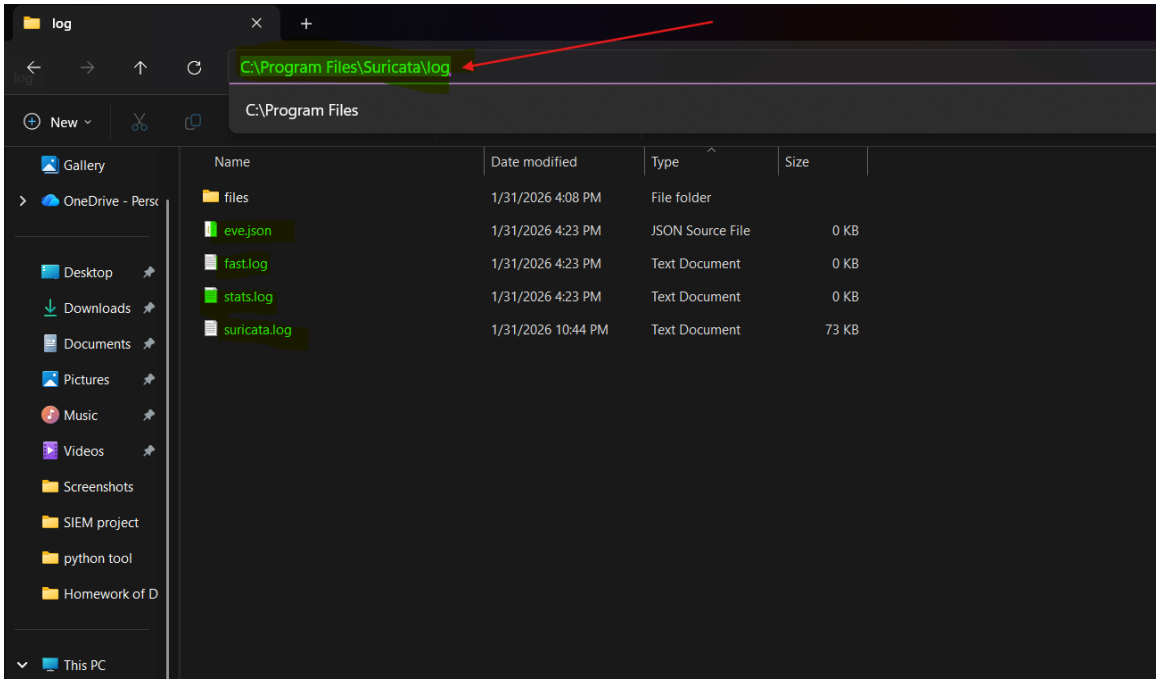


Then in suricata.yaml file add those rules in rule-files:



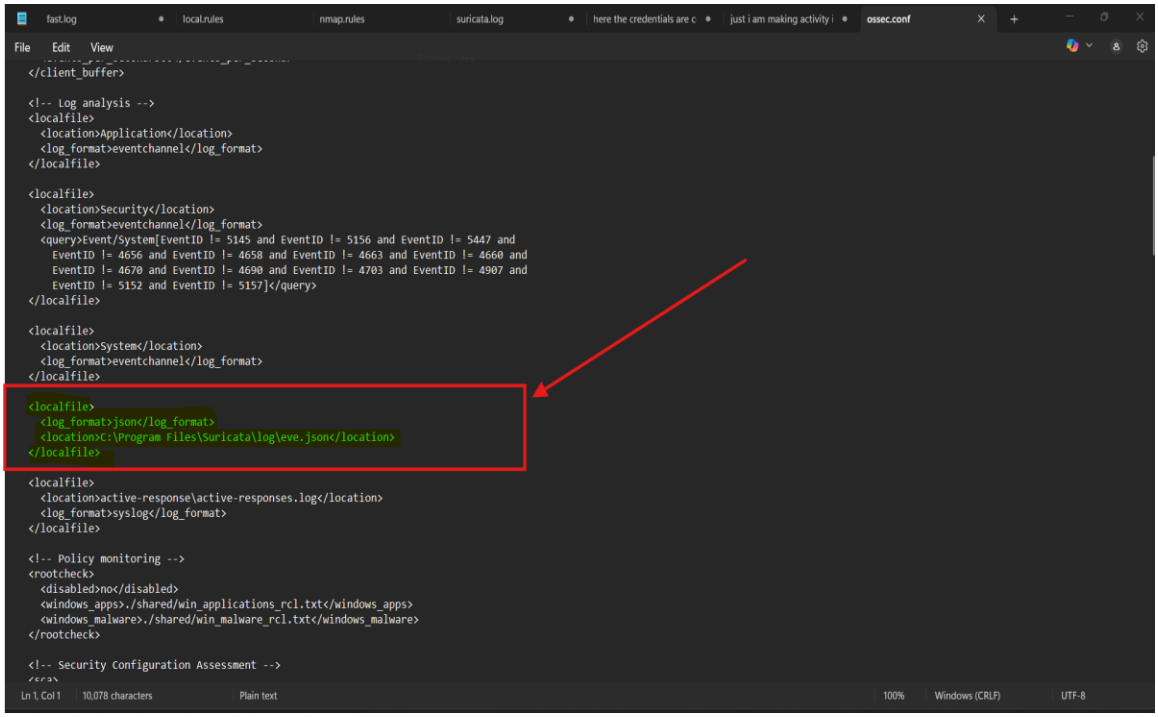
Integrating Suricata logs to Wazuh

In order to integrate Suricata logs in Wazuh we need Wazuh agent to collect Suricata Logs. The logs generated by Suricata is stored in "C:\Program Files\Suricata\log\eve.json"

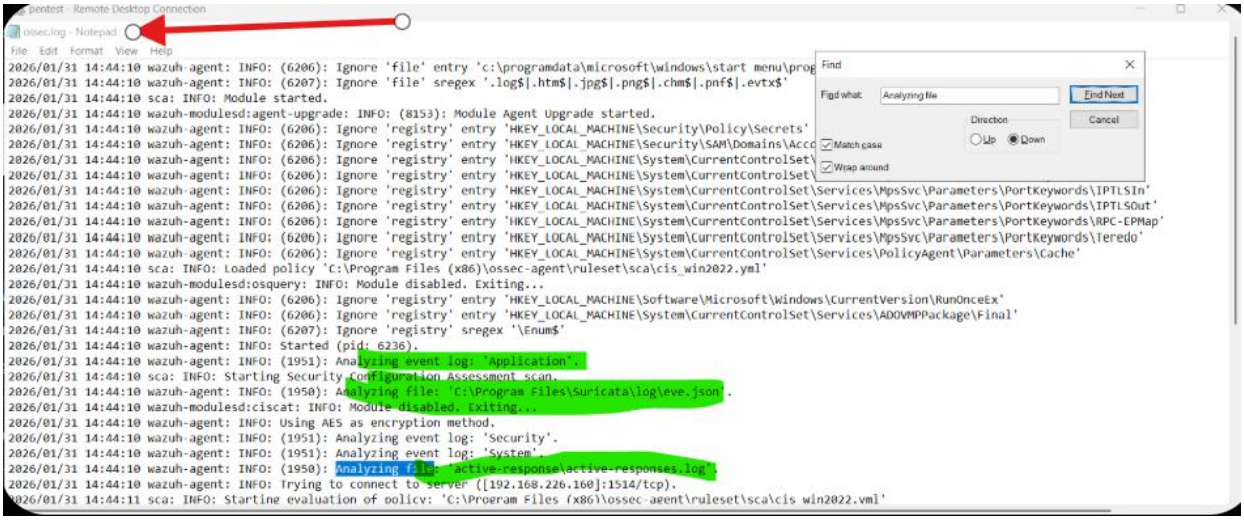


After that access the Wazuh-Agent configuration file ossec.conf from

"C:\Program Files(x86) \ossec-agent" folder. Then In ossec.conf provide the path of Suricata logs to collect from.



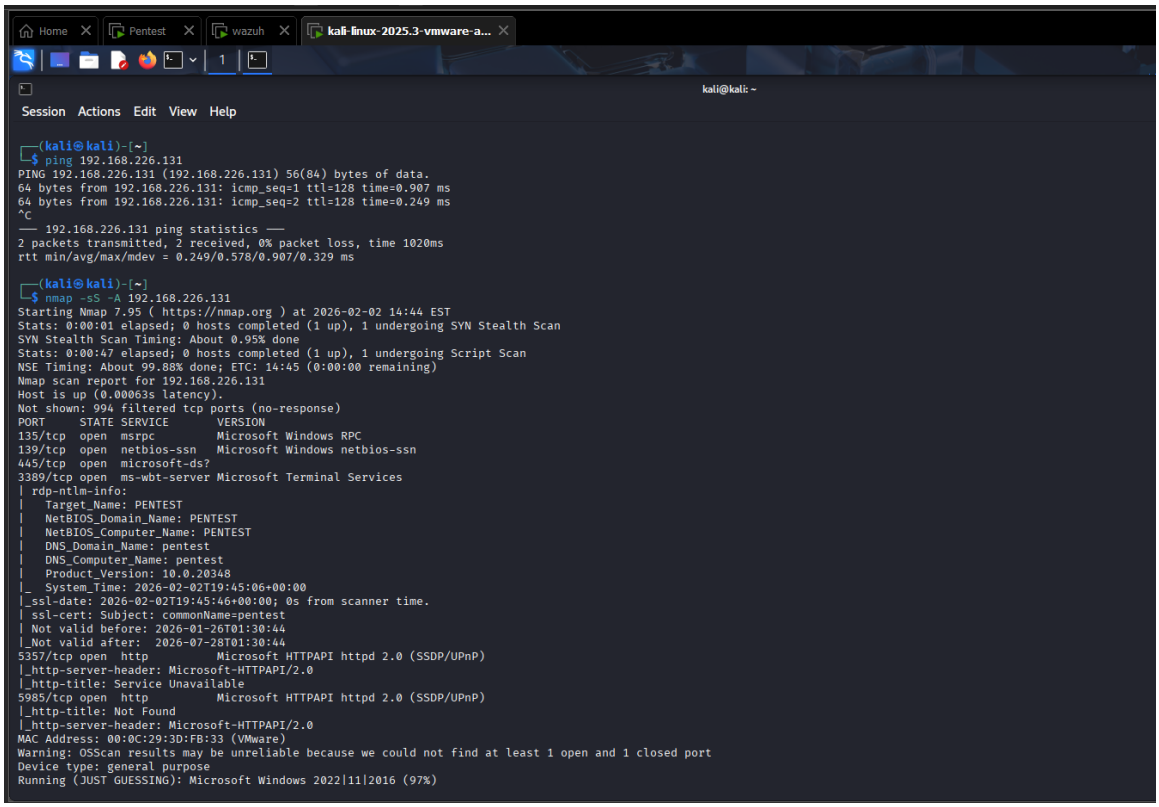
Verify connection of Suricata with Wazuh



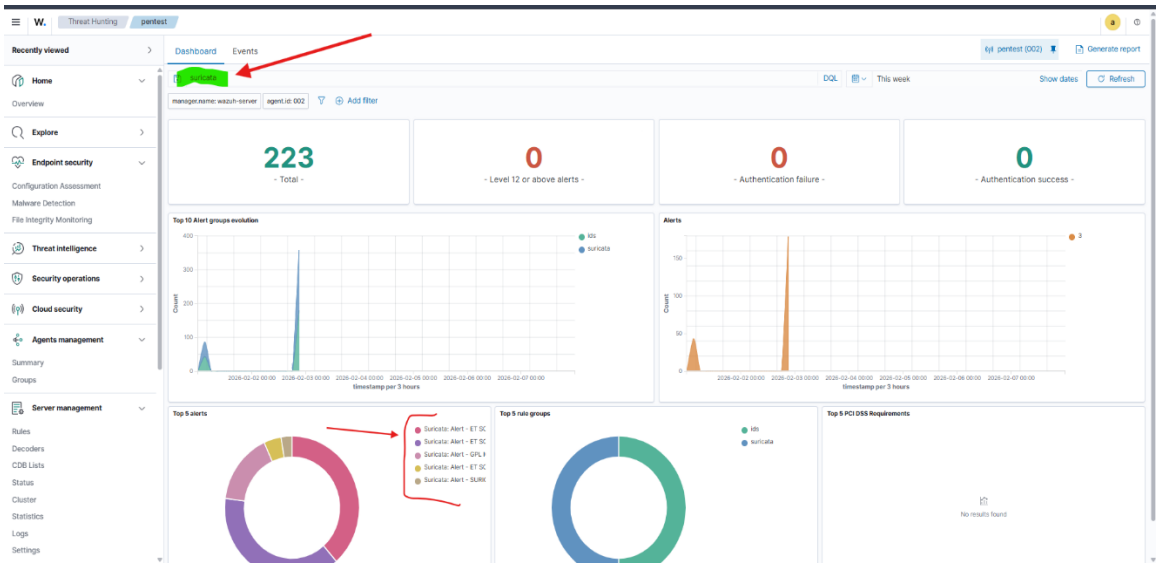
Ensure Wazuh could read the Suricata logs , through following command in powershell:

⇨ Get-Content "C:\Program Files (x86)\ossec-agent\ossec.log" -Tail 50

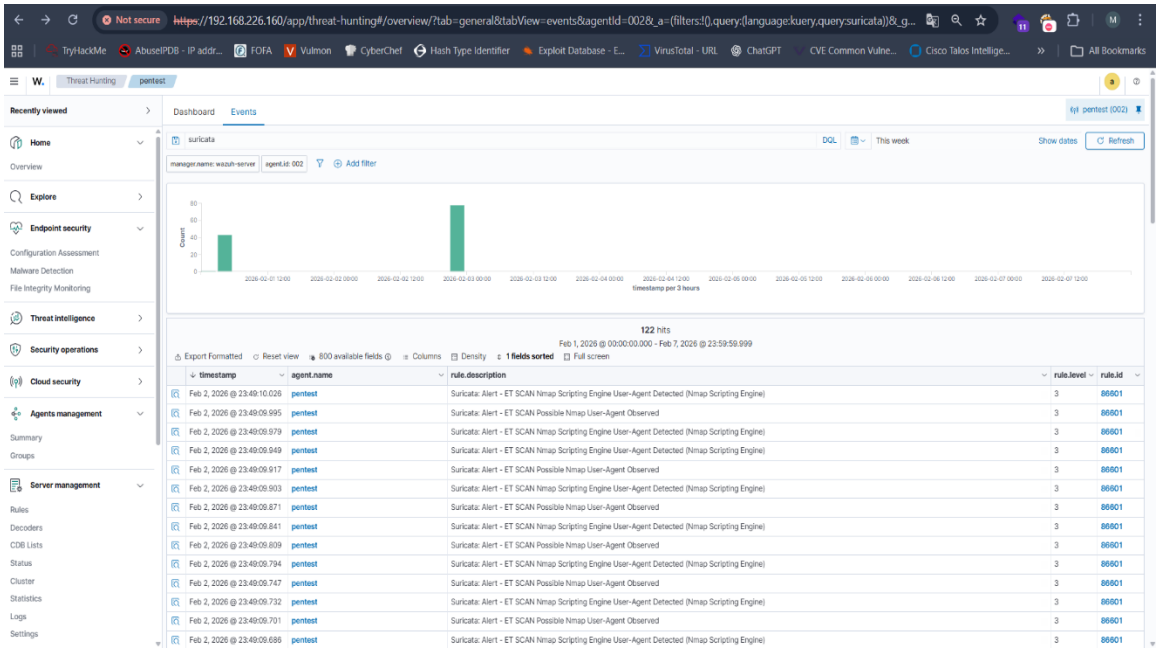
Verify Wazuh-Agent and Suricata IDS with simple information gathering using “NMAP”:



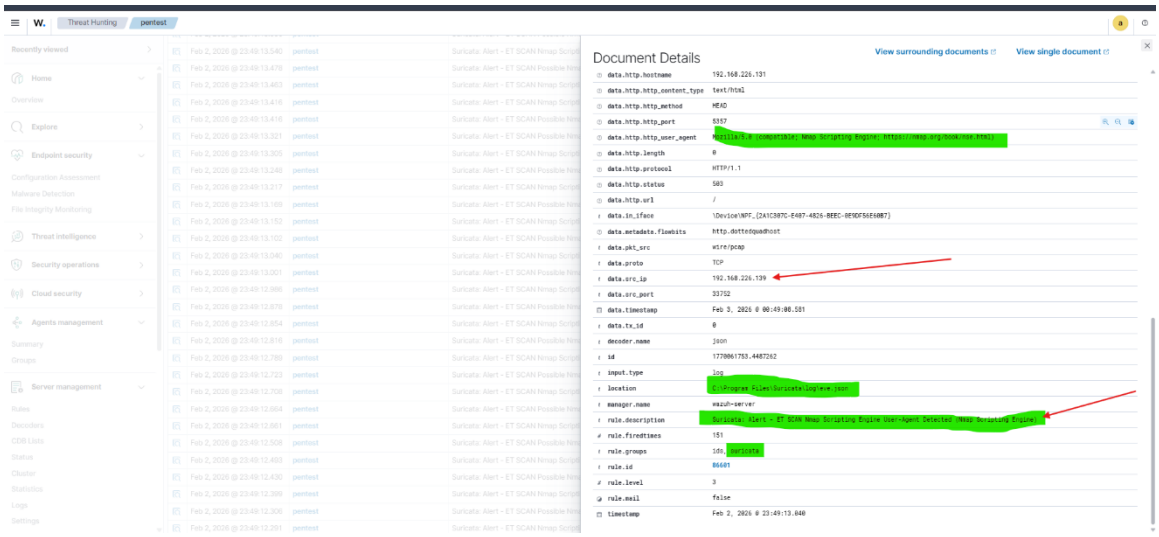
In Wazuh-server Filter the Suricata logs for results:



Show Suricata alerts in Wazuh-dashboard:



document detail about a specific event :



The document detail of suricata alert with Wazuh-Agent gathered information and they show the following :

1. Type of attack according to the rule that is in the Suricata.
2. From where that alert was coming
3. The source IP of this suspicious activity that happened in the system.
4. The alert signature from Wazuh-Agent which was

The Full report in the dashboard of Wazuh-server is down:

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@timestamp	Feb 2, 2026 @ 23:49:13.040
_index	wazuh-alerts-4.x-2026.02.02
agent.id	002
agent.ip	192.168.226.131
agent.name	pentest
data.alert.action	allowed
data.alert.category	Web Application Attack
data.alert.gid	1
data.alert.metadata.confidence	High
data.alert.metadata.created_at	2010_07_30
data.alert.metadata.signature_severity	Informational
data.alert.metadata.updated_at	2024_04_13
data.alert.rev	8
data.alert.severity	1
data.alert.signature	ET SCAN Nmap Scripting Engine User-Agent Detected (Nmap Scripting
data.alert.signature_id	2009358
data.app_proto	http
data.dest_ip	192.168.226.131
data.dest_port	5357
data.direction	to_server
data.event_type	alert
data.flow.bytes_toclient	327
data.flow.bytes_toserver	431
data.flow.dest_ip	192.168.226.131
data.flow.dest_port	5357
data.flow.pkts_toclient	2

data.flow.pkts_toserver	4
data.flow.src_ip	192.168.226.139
data.flow.src_port	33752
data.flow.start	2026-02-02T11:49:08.486280-0900
data.flow_id	1244134072201585.000000
data.http.hostname	192.168.226.131
data.http.http_content_type	text/html
data.http.http_method	HEAD
data.http.http_port	5357
data.http.http_user_agent	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/b
data.http.length	0
data.http.protocol	HTTP/1.1
data.http.status	503
data.http.url	/
data.in_iface	\Device\NPF_{2A1C307C-E407-4826-BEEC-0E9DF56E60B7}
data.metadata.flowbits	http.dottedquadhost
data.pkt_src	wire/pcap
data.proto	TCP
data.src_ip	192.168.226.139
data.src_port	33752
data.timestamp	Feb 3, 2026 @ 00:49:08.581
data.tx_id	0
decoder.name	json
id	1770061753.4487262
input.type	log
location	C:\Program Files\Suricata\log\eve.json

manager.name	wazuh-server
rule.description	Suricata: Alert - ET SCAN Nmap Scripting Engine User-Agent Detecte
rule.firedtimes	151
rule.groups	ids, suricata
rule.id	86601
rule.level	3
rule.mail	false
timestamp	Feb 2, 2026 @ 23:49:13.040